# The empirical recurrence for OEIS A205050, proved by transfer matrix

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**Abstract**

OEIS A205050 counts arrays over $\{0, \dots, 2\}$ under a condition on the number of rightwards and downwards edge increases in every $2 \times 2$ subblock, and carries an empirical recurrence of order 39 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is local to a bounded window of consecutive rows, so the count is a walk count on $S = 803$ states and is $C$-finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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## 1 The conjecture

OEIS A205050 is "Number of (n+1)X4 0..2 arrays with the number of rightwards and downwards edge increases in each 2X2 subblock equal to the number in all its horizontal and vertical neighbors.". It has offset 1 and begins

$$803, 4171, 23720, 182720, 1280452, 10523741, 76554719, 627391509, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 9*a(n-1) +19*a(n-2) -374*a(n-3) +1554*a(n-4) -3313*a(n-5) +2037*a(n-6) +13607*a(n-7) -59774*a(n-8) +133830*a(n-9) -181169*a(n-10) +85068*a(n-11) +288109*a(n-12) -978591*a(n-13) +1883539*a(n-14) -2936585*a(n-15) +4267787*a(n-16) -5889392*a(n-17) +7243591*a(n-18) -7562613*a(n-19) +7078420*a(n-20) -7501425*a(n-21) +10198666*a(n-22) -13740764*a(n-23) +14894742*a(n-24) -12577898*a(n-25) +8938306*a(n-26) -6164409*a(n-27) +4354773*a(n-28) -2816255*a(n-29) +1501042*a(n-30) -672694*a(n-31) +293114*a(n-32) -135385*a(n-33) +58189*a(n-34) -20496*a(n-35) +6012*a(n-36) -1564*a(n-37) +324*a(n-38) -36*a(n-39) for n>43

As of the "Last modified" line on the live entry (Oct 03 2025 11:43:20, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 What is being counted

For a $2 \times 2$ subblock $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ the entry's quantity, the number of rightwards and downwards edge increases, counts how many of the four steps of the cycle $p \to q,\ r \to s,\ p \to r,\ q \to s$ go up in value. Write $\sigma$ for it. The subblocks of an $(n+1) \times 4$ array over $\{0, \dots, 2\}$ form a grid of $n$ rows and 3 columns and $\sigma$ labels that grid.

The reading is not a guess. Over 0..1 the $2 \times 3$ arrays whose two subblocks have equal clockwise counts number 40; over 0..2 the $2 \times 2$ arrays of clockwise count 2 number 30; the $3 \times 2$ binary arrays whose two subblocks have unequal rightwards-and-downwards counts number 40. Those are the first terms the corresponding entries publish, and each was computed by direct enumeration before any transfer matrix was built.

The condition relates NEIGHBOURING subblocks: two subblocks adjacent in the grid, horizontally or vertically, must carry equal values of $\sigma$.

## 3 The count is a walk count

A subblock's value needs two consecutive array rows, and the vertical comparison needs the two subblock rows that three consecutive rows produce. So the state is the PAIR of consecutive array rows: the horizontal comparisons decide which pairs are admissible at all, the vertical ones decide which pair may follow which. A step appends one array row, turning $(r, s)$ into $(s, t)$ and settling every comparison between the two subblock rows at once. An $(n + 1)$-row array is a walk of $n - 1$ steps, so

$$a(n) \;=\; \iota^{\top} M^{\,n-1} \tau, \qquad \iota = \tau = (1, \dots, 1)^{\top},$$

on the $S = 803$ admissible pairs.

At $n = 1$ the array has two rows, so the grid is a single row of 3 subblocks; the entry's first term is 803, which the model reproduces along with every later term. That is the cheapest check that the grid has been read with the right shape. In particular $a$ is $C$-finite. The alignment of the walk index with the entry's own $n$ was checked against its published terms rather than assumed.

## 4 The proof

**Lemma 1.** *Let $q(t) = t^{39} - \sum_i c_i t^{39-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n - i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index $j$, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large $j$, and it is enough to examine $j < S$.*

*Proof.* Factor $M^j$ out of $M^{j+39} - \sum_i c_i M^{j+39-i}$. For the bound, $M^S$ is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of $M$; $S$ consecutive zeros force every later one, and the last nonzero $u_j$ pins the threshold exactly. □

**Theorem 2.**

$$a(n) \;=\; 9\,a(n{-}1){+}19\,a(n{-}2){-}374\,a(n{-}3){+}1554\,a(n{-}4){-}3313\,a(n{-}5){+}2037\,a(n{-}6){+}13607\,a(n{-}7){-}59774\,a($$

*for every $n > 43$.*

*Proof.* The vector $w = q(M)\tau$ was formed by 39 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 43 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger $j$ at once. □

## 5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

## References


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